

Sarath Kumar Senthilkumar Kavitha

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LinkedIn | GitHub | Portfolio

Profile

Robotics engineer with hands-on experience designing, developing and integrating robotic and automation solutions from prototype through to real-world deployment on industrial and mobile systems. Practical background in control system design and tuning (cascade PID, closed-loop feedback control), multi-sensor data fusion (LiDAR, depth camera, IMU), and real-time embedded systems on Linux (NVIDIA Jetson Xavier, Raspberry Pi). Delivered a closed-loop, edge-to-actuation AI system into an existing industrial inspection workflow, and taken control algorithms end-to-end, from simulation through to live testing on real hardware, including flight tests on a real drone and deployment on a live mobile robot platform. Proficient in C++ and Python, with hands-on computer vision experience (OpenCV, YOLO) and a track record of validating systems in simulation before hardware deployment.

Technical Skills

Control Systems:	Cascade PID control (inner/outer loop design), closed-loop feedback control, control gain tuning & validation, sim-to-real transfer
Sensing & Data Fusion:	LiDAR, depth camera, IMU, wheel encoders, multi-sensor data fusion, SLAM
Real-Time & Embedded Systems:	Linux-based real-time systems, NVIDIA Jetson Xavier, Raspberry Pi, Arduino, IoT
Computer Vision & AI:	OpenCV, YOLO, TensorFlow, CNN, Deep Learning, Computer Vision, Reinforcement Learning
Robotics Frameworks:	ROS1/ROS2, Gazebo, RViz, Nav2, MoveIt2, Path Planning
Languages:	C, C++, Python, MATLAB, JavaScript, Java, SQL
Protocols:	UART, I2C, SPI, MQTT, HTTP
Tools & Practices:	Linux, Git, Docker, design & code review, unit/integration testing, technical documentation, debugging & troubleshooting

Professional Experience

Robotics Engineer Intern <i>Bimental Pvt. Ltd. (VIT Contractual Project)</i>	<i>Jan 2025 – Aug 2025</i> Chennai, India
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- Analysed the inspection line's automation needs and built a real-time surface defect detection system from the ground up, running TensorFlow and YOLO inference on an NVIDIA Jetson Xavier embedded computer. Achieved 85% detection accuracy in an industrial production environment with no cloud connectivity.
- Redesigned the preprocessing pipeline and CNN feature extraction approach, lifting detection precision by 89%.
- Integrated the AI system with a robotic spray actuation mechanism into the client's existing industrial inspection workflow, delivering a closed-loop, edge-to-actuation solution on time and to spec. Collaborated with three engineering teams to bring it into production.

ROS2 Programming Intern <i>Mafkin Robotics</i>	<i>Jun 2024 – Aug 2024</i> Chennai, India
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- Programmed the localization and motion control systems for an autonomous ship-cleaning robot using ROS2 on a Linux-based real-time platform, integrating with hardware sensors on a live mobile system. Fused data from a camera, IMU and wheel encoders to achieve real-time mapping, obstacle avoidance and smooth path following.
- Supported troubleshooting from prototype to final system: debugged and tuned localization drift and control instability issues during real-world testing, iterating on sensor fusion parameters to hit the precision localization targets the project required.
- Documented the software architecture and integration workflow for the localization stack, building sensor integration experience that transfers directly to fleet-scale mobile robotics and ADAS development.

RPA Developer Intern <i>Tata Consultancy Services</i>	<i>Aug 2023 – Nov 2023</i> Chennai, India
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- Designed and deployed 15 automated business process workflows using UiPath; earned the UiPath Certified RPA Developer credential during the placement.

Projects

Leo Rover Autonomous Navigation and Manipulation

May 2026

MSc Robotics, University of Manchester

- Built a Leo Rover system integrating a robotic arm, Raspberry Pi, depth camera and 2D-LiDAR for fully autonomous navigation and manipulation on real hardware. Implemented ROS2-based frontier exploration for autonomous environment mapping, achieving localisation accuracy within ± 12 cm with real-time path planning and obstacle avoidance.
- Developed a computer vision pipeline using the depth camera to detect colored blocks in real time, with manipulator control logic to autonomously pick up blocks and place them in designated positions. Integrated and debugged sensor fusion across LiDAR, depth camera and IMU for reliable indoor perception, validating first in Gazebo before hardware deployment.

Cascade PID Drone Control

Mar 2026

- Designed and implemented a cascade PID control system for autonomous drone stabilisation, with an inner loop regulating angular rate and an outer loop controlling attitude and altitude, a linear control architecture tuned through iterative simulation testing before deploying and validating the controller on a real drone in live flight tests.
- Validated the full quadrotor dynamics model in Python simulation, then transferred the controller to hardware, achieving stable hover and responsive setpoint tracking. Debugged and demonstrated robust recovery from external perturbations in real-time flight, confirming the sim-to-real transfer of the control design.

Robotic Automation for Fault Detection in Metal Sheets

Aug 2025

- Designed and validated a complete edge AI pipeline: OpenCV preprocessing, YOLO object detection and TensorFlow classification running on NVIDIA Jetson Xavier, achieving 89% defect identification accuracy and improving throughput by 40%.
- Connected the inference output to a physical robotic spray actuator, demonstrating closed-loop embedded AI system design from sensor to actuation in an industrial automation setting.

Nuclear Environment Automation System

Feb 2025

- Designed an autonomous robot for operation in hazardous nuclear environments, using Visual SLAM for real-time mapping and localization. Validated navigation accuracy and sensor performance through thorough Gazebo simulation testing before any hardware work.
- Embedded depth cameras for perception and attached a robotic arm for autonomous task execution, combining manipulation and navigation in a single system.

Education

MSc Robotics

Sep 2025 – Present

The University of Manchester, Manchester, UK

Modules: Robotic Systems, Computer Vision, Reinforcement Learning, Autonomous Mobile Robots, Robotic Manipulators, Cognitive Robotics, Software for Robotics

BTech Computer Science Engineering with AI and Robotics

Sep 2021 – Jul 2025

Vellore Institute of Technology, Chennai, India

CGPA: 9.07/10.0 (First-Class Honours)

Modules: Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Natural Language Processing, Sensors and Actuators, Robotics Kinematics, Robot Operating System (ROS), Cognitive Robotics, Robotics Perception

Awards and Certifications

- UiPath Certified RPA Developer Foundation, October 2023
- Google Cloud Computing Foundation (NPTEL), November 2023
- 1st Place at CSI One-Day Product Development Hackathon 2023, Computer Society of India
- Stage 3 Finalist at e-Yantra Robotics Competition 2024, IIT Bombay — top 100 teams globally